When we start Elasticsearch on a machine, it binds to the localhost by default. We can change this binding to a specific network address by changing `network.host` and `network.port` (as well as `transport.port` for node-to-node networking) if needed.

Changing these settings on a farm of computers is a pain, especially when you need to set up a cluster with hundreds of nodes. Be sure you have housekeeping scripts handy to alleviate this nuisance. One ideal way is to create configurations in a central folder and point the `ES_PATH_CONF` variable to those settings. (We can also use Ansible, Azure Pipelines, GitOps, etc., for such purposes.) Exporting this variable lets Elasticsearch choose the configuration from this directory.

Coming back to setting up the network properties, we can use special values in the config file to set the network host rather than configuring the host manually. Setting the `network.host` property to `_local_` lets Elasticsearch set its address automatically. This sets the loopback address (127.0.0.1) as the network host. The `_local_` special value is the default for the `network.host` attribute. My suggestion is to leave it untouched.

There's also a `_site_` value that sets the `network.host` attribute to site-local addresses (192.168.0.1, for example). We can set `network.host`, defaulting to these to special values by setting `network.host: [_local_, _site_]` in the configuration.

14.3 Shard sizing

One question that always arises when we talk about shards is sizing. Look at shard sizing in detail, especially the memory footprint we need to consider and other variables, to better understand this topic. For argument's sake, we focus on a use case with one index on a five-node cluster and then find the shard sizing with multi-indexes on the cluster.

14.3.1 Setting up a single index

Let's say we have a 5-node cluster with one index comprising 10 primary shards and 2 replicas. This means each shard has 2 replicas, and the total number of shards is 30 for both primary nodes and replicas. Figure 14.6 shows this configuration.

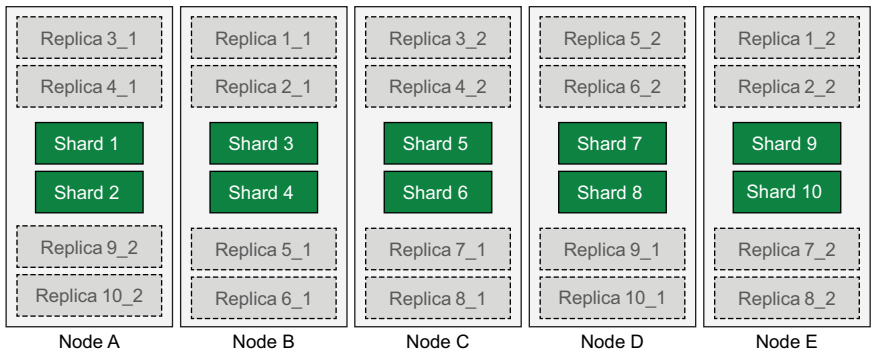


Figure 14.6 Shards and replicas spread across a multi-node cluster